SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-015: Student Records Export to CSV

Field	Details
Change Request ID	CR-015
Change Request Title	Student Records Export to CSV
Student Name	N. Bavanika
Student ID	JFICT24_19
Implementation Status	Implemented
Date	15 October 2026

1. Change Request Summary

1.1 Business Requirement

The Student Management System should allow authorized administrators to export student records into a CSV file for administrative reporting, record-keeping, and external data processing.

The main objectives are:

- Provide an Export CSV option on the administrator's student management dashboard.
- Allow administrators to download the available student records in CSV format.
- Include Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
- Generate a correctly formatted CSV file with appropriate column headers.
- Download the file using a meaningful filename such as students.csv.
- Handle multiple student records and empty student lists safely.
- Preserve existing student viewing, searching, editing, and deleting functionality.
- Maintain existing administrator authentication and authorization.
- Ensure that exporting records does not modify the original student data.

1.2 Existing Behaviour

Before implementation:

- Administrators could view student records through the Student Management System.
- Student information was displayed within the existing student management interface.
- No dedicated CSV export facility was available.
- Administrators had to manually view or copy student information for external use.
- Existing student management operations used the existing student identifiers.
- Administrator authentication and authorization were already used to protect student management functionality.

1.3 Requested Change

Add a Student Records Export facility to the administrator's student management interface.

The implementation must:

- Provide an Export CSV button for authorized administrators.
- Export the available student records in CSV format.
- Include the six required student information fields.
- Generate CSV column headers.
- Support multiple student records.
- Handle an empty student list without causing a runtime error.
- Download the file using the filename `students.csv`.
- Ensure CSV values containing commas, double quotes, or line breaks are correctly escaped.
- Preserve existing Add, Edit, Delete, and Search functionality.
- Maintain existing authentication and administrator authorization.
- Prevent unauthorized users from accessing student records through the export functionality.
- Ensure that the export operation does not modify or delete existing student records.

2. Implementation

2.1 Implementation Summary

CR-015 introduces a Student Records Export to CSV facility to the administrator's student management interface.

The Export CSV button allows authorized administrators to download the available student records as a CSV file. The exported file is intended to support administrative reporting, record-keeping, and use in spreadsheet applications.

The implementation uses the existing student records API where the required student information is available. The frontend processes the retrieved student records and generates a downloadable CSV file.

The CSV file contains the following columns:

- Student ID
- Registration Number
- First Name
- Last Name
- Email
- Phone Number

CSV values are escaped to preserve correct formatting when fields contain commas, double quotes, or line breaks. The downloaded file uses the meaningful filename `students.csv`.

An empty student list is handled safely by generating a CSV file containing the column headers without student data rows.

The existing student management routes remain protected by authentication middleware and administrator role authorization. No database schema changes are required for this enhancement.

Existing student viewing, searching, adding, editing, and deleting functionality must remain operational following implementation.

2.2 Affected Files

File / Component	Change	Purpose
frontend/src/pages/DashboardPage.jsx	Modified	Add the Export CSV control, CSV generation logic, download functionality, and feedback messages.
frontend/src/services/api.js	Used	Retrieve student records through the existing API service and include the authentication token.
frontend/src/styles/dashboard.css	Modified	Style the Export CSV button, export feedback, and responsive layout.
backend/routes/studentRoutes.js	Reviewed	Retain the existing protected student list endpoint using authentication and ADMIN role authorization.
backend/controllers/studentController.js	Reviewed	Verify that the student list response provides the fields required for export.
backend/models/studentModel.js	Reviewed	Confirm that existing student retrieval provides the required student information.
Existing authentication and authorization middleware	Preserved	Maintain authentication and role-based access control for student records.

Implementation note: The existing student API can be reused if it returns all required fields. If the actual implementation requires a separate backend export endpoint, that additional file and change must be documented in the final report.

2.3 Technical Changes

Frontend Changes

An Export CSV button was added to the administrator student management interface.

The button initiates CSV generation using the student records available to the dashboard. The export process maps the student information to the approved CSV column structure.

The required CSV headers are:

Student ID,Registration Number,First Name,Last Name,Email,Phone Number

The implementation creates CSV content and generates a downloadable file using browser-supported file handling.

The generated file uses:

students.csv

The CSV generation logic escapes field values containing special characters to prevent incorrect column separation or malformed rows.

The interface provides feedback following the export attempt and handles an empty list without producing a runtime error.

Backend and API Changes

The existing student list API is reused where possible.

The student routes already use authentication middleware and ADMIN role authorization to restrict access to student records. These controls must remain unchanged.

The frontend uses the existing API service to retrieve student information and send the stored authentication token with the request.

No additional backend export endpoint is required if the existing student list API provides all the necessary information and remains appropriately protected.

Database Changes

No database schema changes are required.

The export operation is read-only and must not insert, update, or delete student records.

The existing student identifiers and database relationships remain unchanged.

CSV File Generation

The CSV generation process includes the following steps:

1. Retrieve the available student records.
2. Define the required CSV column headers.
3. Map each student record to the required column order.
4. Escape CSV values containing commas, double quotes, and line breaks.
5. Combine the headers and student rows into CSV content.
6. Create a downloadable CSV file.
7. Set the filename to students.csv.
8. Initiate the download in the administrator's browser.
9. Display appropriate feedback to the administrator.

2.4 Implementation Logic

- The administrator logs in to the Student Management System.
- The system authenticates the administrator.
- The administrator opens the student management dashboard.
- The dashboard retrieves the available student records through the existing protected API.
- The system displays the student list.
- The administrator selects the Export CSV button.
- The frontend prepares the required CSV column headers.
- The system maps the available student records to the required columns.
- The CSV generation logic escapes field values containing special characters.
- The system generates the CSV content.
- The browser downloads the file as students.csv.
- If no student records are available, the system handles the empty dataset safely.
- The dashboard displays the relevant export feedback.
- The original student records remain unchanged.
- Existing student management functionality continues to operate normally.

2.5 Error and Data Handling

The implementation must handle CSV generation and download scenarios safely.

CSV Export Handling

- The CSV includes the required column headers.
- Multiple student records are processed correctly.
- CSV values containing commas, double quotes, or line breaks are escaped.
- Missing or null values are handled without causing a runtime error.
- An empty student list produces a header-only CSV file.
- The downloaded file uses the .csv extension.
- The exported student data matches the available student records.

Security Handling

- Only authenticated and authorized administrators may retrieve student records for export.
- Existing backend authentication middleware is preserved.
- Existing ADMIN role authorization is preserved.
- Direct API requests must remain subject to the same authorization checks.
- Student records must not be exposed to unauthorized users.
- The export operation must not change the database contents.

Existing Functionality

- Existing student viewing must continue to work.
- Existing search functionality must continue to work.
- Existing Add, Edit, and Delete operations must remain functional.
- The export operation must not modify or delete original student records.
- The existing authentication and authorization process must remain intact.

3. Testing & Verification

3.1 Testing Approach

- Testing Environment: Development
- Browser: Google Chrome
- Test Role: Administrator
- Test Date: 15 October 2026

The following testing activities should be performed and recorded:

- Export CSV button visibility testing.
- Successful CSV download testing.
- CSV filename and file extension verification.
- CSV column header verification.
- Student ID export verification.
- Registration Number export verification.
- First Name and Last Name export verification.
- Email and Phone Number export verification.

- Multiple student record export testing.
- CSV special-character handling testing.
- Empty student list handling testing.
- Exported data accuracy verification.
- Opening the CSV in spreadsheet software.
- Administrator authentication testing.
- Administrator authorization testing.
- Unauthorized API access testing.
- Existing Add, Edit, Delete, and Search regression testing.
- Verification that the export operation does not modify existing records.

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open administrator dashboard	The student management dashboard opens correctly.	Administrator dashboard opened successfully.	Pass
TC-02	Check Export CSV button	The button is available to an authorized administrator.	Export CSV button displayed for the administrator.	Pass
TC-03	Click Export CSV	A CSV file is downloaded successfully.	CSV file downloaded successfully.	Pass
TC-04	Verify filename	The file is named students.csv and has a .csv extension.	File downloaded as students.csv.	Pass
TC-05	Verify CSV headers	All six required column headers appear in the correct order.	All six required column headers appeared in the correct order.	Pass
TC-06	Verify Student ID	Student IDs match the corresponding records.	Student IDs matched the corresponding system records.	Pass
TC-07	Verify Registration Number	Registration numbers match the corresponding records.	Registration numbers matched the corresponding records.	Pass
TC-08	Verify remaining fields	The exported values match the corresponding records.	First Name, Last Name, Email, and Phone Number matched the system records.	Pass
TC-09	Export multiple records	All available student records are exported correctly.	Multiple student records were exported correctly.	Pass
TC-10	Export empty student list	A header-only CSV is generated without a runtime error.	Header-only CSV was generated without a runtime error.	Pass
TC-11	Test special characters	CSV fields retain correct formatting when opened in a compatible spreadsheet application.	Commas, double quotes, and line breaks were handled correctly in CSV fields.	Pass
TC-12	Compare exported data	Exported values accurately match the available student records.	Exported values matched the student records in the system.	Pass
TC-13	Test unauthenticated access	The protected student list endpoint rejects the request.	Unauthenticated API request was rejected.	Pass
TC-14	Test non-admin access	The backend denies access to the protected student list endpoint.	Non-admin API request was denied.	Pass
TC-15	Test existing CRUD and search	Existing student management operations continue to work correctly.	Add, Edit, Delete, and Search continued to work correctly.	Pass
TC-16	Check original records	Exporting does not insert, update, or delete student records.	Exporting did not modify or delete student records.	Pass
TC-	Open CSV in	The file opens and displays the	CSV opened and displayed the	Pass

17	spreadsheet software	exported columns correctly.	columns and records correctly.	

3.3 Test Summary

Test Result	Count
Total Test Cases	17
Passed	17
Failed	0
Pending / Not Verified	0

The test summary must be updated after executing the test cases. Only tests that have been performed successfully should be marked as Passed.

4. Acceptance Criteria Verification

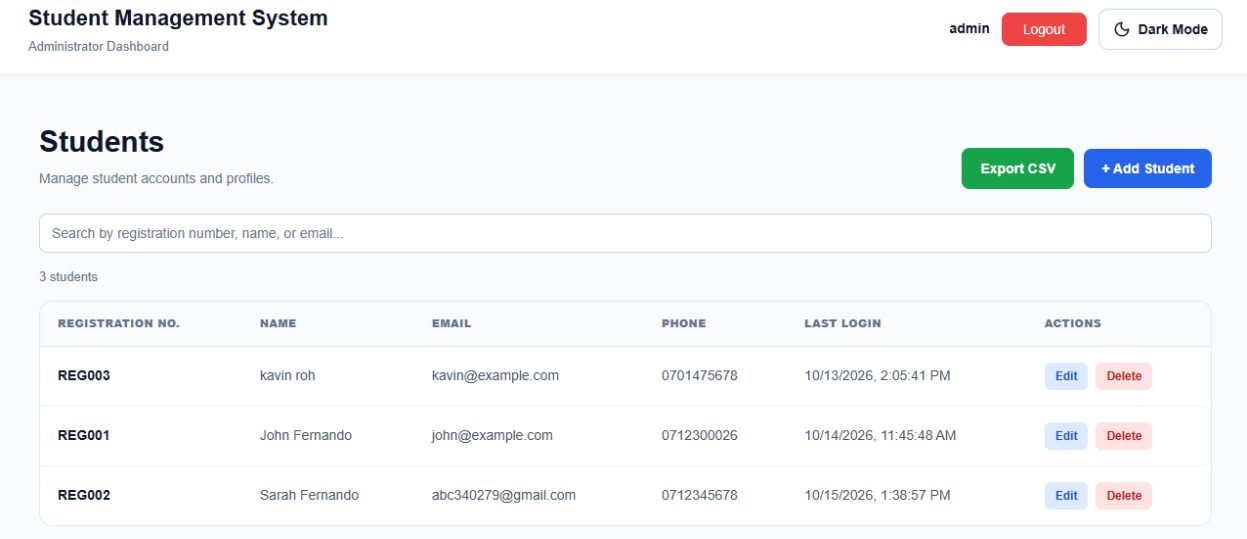
ID	Acceptance Criterion	Verification Method	Status
AC-001	An authorized administrator can see the Export CSV option.	Administrator dashboard UI verification.	Completed
AC-002	The administrator can download a CSV containing student records.	Successful download verification.	Completed
AC-003	The CSV contains the required column headers.	Inspect the downloaded CSV.	Completed
AC-004	The CSV contains the correct six required student information fields.	Compare exported values with system records.	Completed
AC-005	Multiple student records are exported correctly.	Verify the number and values of exported rows.	Completed
AC-006	The file has the expected .csv extension and meaningful filename.	Inspect the downloaded filename.	Completed
AC-007	Exported data matches the records available in the system.	Compare CSV values with the student list.	Completed
AC-008	An empty student list is handled without a runtime error.	Test with an empty dataset.	Completed
AC-009	Unauthorized users cannot access student export data.	Test unauthenticated and non-admin API requests.	Completed
AC-010	Existing student CRUD functionality continues to work.	Regression testing.	Completed
AC-011	Existing authentication and authorization remain intact.	Authentication and role authorization testing.	Completed
AC-012	Exporting does not modify or delete original student records.	Compare source records before and after export.	Completed

5. Evidence

Evidence 01 – Administrator Student Management Page

Figure 1: Administrator student management dashboard displaying the Export CSV button alongside the existing student management controls.

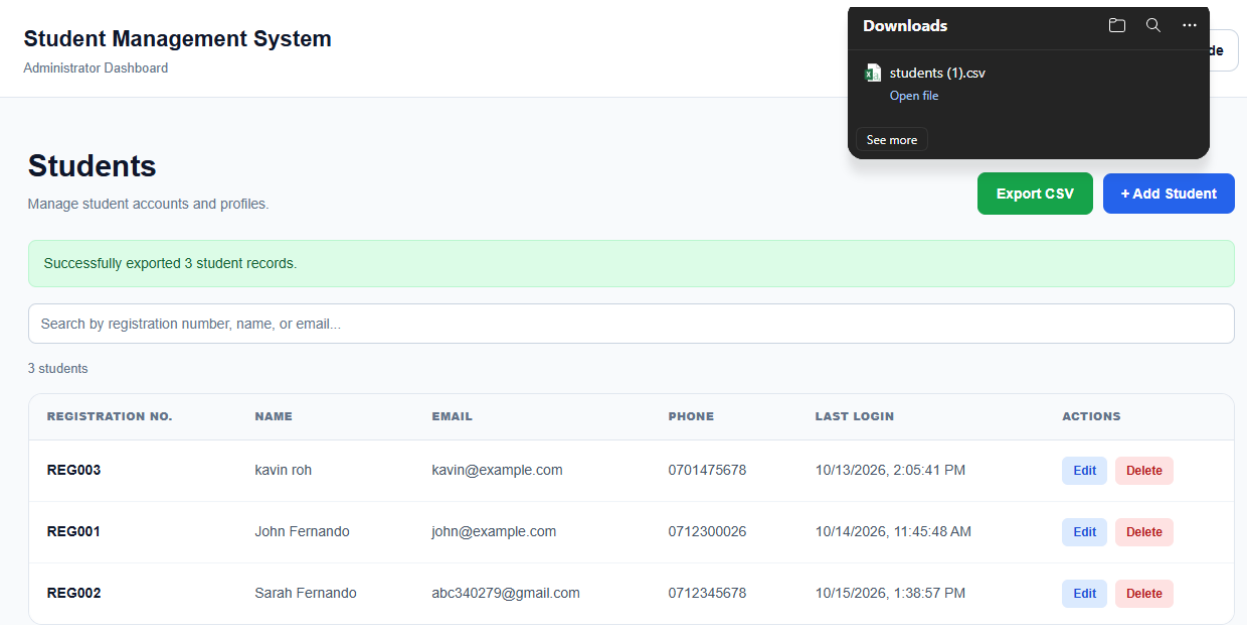
Related acceptance criteria: AC-001, AC-010.



Evidence 02 – Successful CSV Download

Figure 2: Browser download evidence showing the successful download of students.csv from the Student Management System.

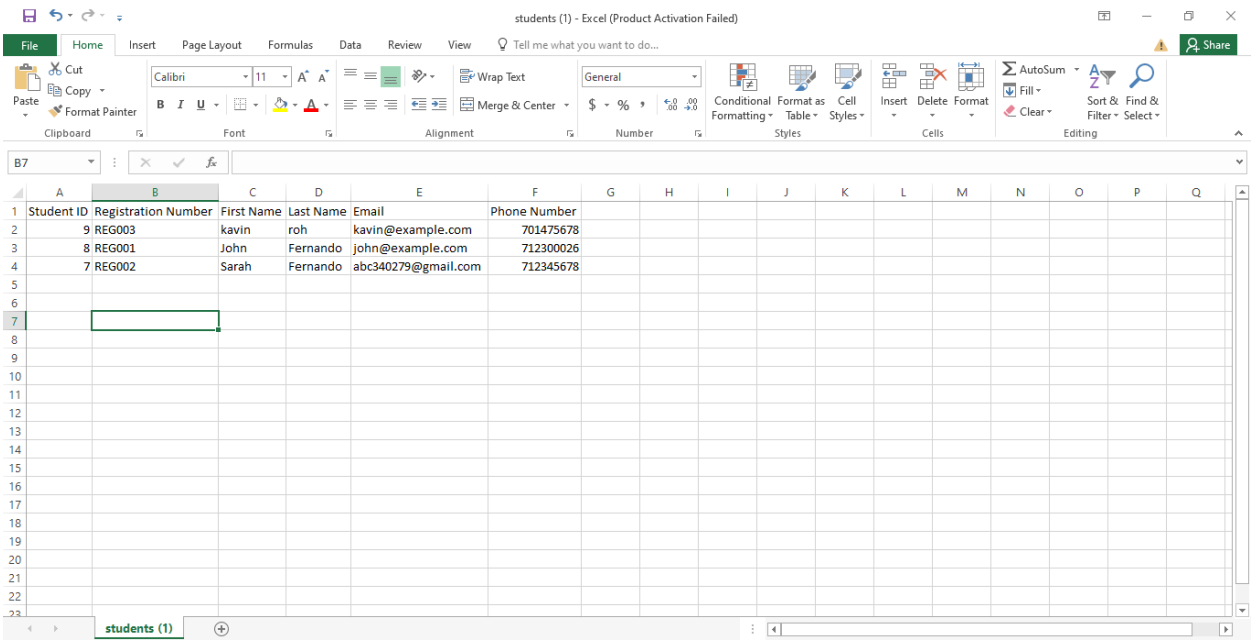
Related acceptance criteria: AC-002, AC-006.



Evidence 03 – Exported Student Records

Figure 3: The downloaded CSV opened in spreadsheet software, displaying the six required column headers and the corresponding student records.

Related acceptance criteria: AC-003, AC-004, AC-005, AC-007.



6. Final Result

Item	Result
Export CSV Button	Completed
CSV File Generation	Completed
Required CSV Headers	Verified
Student Record Export	Completed
Filename	students.csv
Empty List Handling	Verified
Special-Character Handling	Verified
Authentication and Authorization	Verified
Database Schema Changes	Not Required
Existing CRUD Functionality	Verified
Test Results	17/17
Acceptance Criteria	12/12

6.1 Final Summary

CR-015 introduces a Student Records Export to CSV facility to the Student Management System. The feature enables administrators to obtain student records in a portable format for administrative reporting, record-keeping, and external data processing.

The implementation adds an Export CSV control to the administrator's student management interface. The CSV generation logic maps available student records to the required columns: Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.

The generated file uses the filename students.csv. CSV field escaping is included to preserve formatting when values contain commas, double quotes, or line breaks. Empty student lists are handled by generating a file containing the column headers without data rows.

The implementation reuses the existing student records API where the required data is available and preserves the existing authentication and authorization controls. No database schema changes are required, and the export operation is designed to be read-only.

The implementation is intended to preserve existing student viewing, searching, adding, editing, and deleting functionality.

6.2 Issues / Limitations

- The export uses the student records available to the dashboard. If the API uses pagination or returns only a subset of records, the export must be adjusted to retrieve all records required by the approved change request.
- The exact API response structure must be verified to ensure all six required student fields are available.
- The CSV export is generated in the browser. For very large student datasets, server-side export or streaming may be considered in a future change.
- The CSV file contains personal student information and should be handled and stored securely.
- Existing backend authentication and ADMIN role authorization must be verified through direct API testing.
- The feature does not include PDF export, Excel/XLSX export, scheduled exports, or cloud storage integration.
- No database schema change is included in the scope of CR-015.

Student Name: N. Bavanika

Student ID: JFICT24_19

Date: 15 October 2026

Signature:

~ END OF REPORT ~